SIDDHARTH HIRAOU

Software Engineer / AI Software Engineer

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SUMMARY

Software Engineer / AI Engineer with around 3 years of experience building Python applications, Generative AI features, and data processing systems. Hands-on in developing RAG pipelines, integrating vector databases and Hugging Face models, and deploying containerized Python services across AWS and Kubernetes environments. Experienced in building automated ETL pipelines, writing optimized SQL queries, integrating SQL/NoSQL databases, and implementing Git-based CI/CD workflows within Agile software development teams.

SKILLS

Languages: Python, SQL, C++

AI & Machine Learning: Generative AI, RAG (Retrieval-Augmented Generation), LLMs (OpenAI APIs), LangChain, Hugging Face Transformers, PyTorch, TensorFlow, Scikit-learn, Vector Databases, Prompt Engineering

Backend & Data Engineering: REST APIs, FastAPI/Flask, Microservices, ETL Pipelines, Pandas, NumPy, Data Wrangling, Asynchronous Processing

Databases: SQL (PostgreSQL, MySQL), NoSQL (MongoDB), Vector DBs (Pinecone/Chroma/FAISS)

Cloud, DevOps & Tools: AWS, Docker, Kubernetes, Git, CI/CD Pipelines, Pytest, Linux, Agile/Scrum

EXPERIENCE

Avnet, NY | Software Engineer / AI Software Engineer **Jan 2026 – Present**

- Developed and maintained Python-based software applications that integrated LLMs, machine learning models, external APIs, and cloud services to support AI-enabled business workflows.
- Built Generative AI applications using Python, LangChain, OpenAI APIs, and Retrieval-Augmented Generation (RAG), integrating vector databases to provide context-aware search and document-based responses.
- Designed and developed Python services and REST APIs to connect AI capabilities with application components, databases, and cloud services, supporting scalable and maintainable production workflows.
- Developed document-processing and NLP features using Hugging Face Transformers, PyTorch, and TensorFlow to support text classification, information extraction, and AI-powered application functionality.
- Integrated SQL and NoSQL databases with Python applications to manage application data, processed documents, search data, and AI-generated results while supporting reliable data retrieval.
- Containerized Python and AI applications using Docker and supported deployment through Kubernetes and AWS services, maintaining consistent development and production environments.
- Implemented Git-based CI/CD workflows to automate application builds, testing, and deployments, improving release consistency across cloud environments.

Siemens, India | Software Engineer **Sep 2022 – Jul 2024**

- Designed and developed modular Python microservices to implement business logic and integrate application workflows with enterprise systems.
- Built reusable Python modules for data validation, transformation, and stream processing, making common processing logic easier to maintain across services.
- Developed RESTful APIs with request validation, error handling, and structured response processing to support communication between internal applications and services.
- Improved relational database performance by developing and tuning SQL queries, stored procedures, and database views used by application workflows and reporting processes.
- Created unit tests with pytest, investigated application issues, and fixed production defects to improve code reliability and reduce recurring software issues.
- Worked with Product, Architecture, and QA teams to understand requirements, define technical approaches, and review code changes before release.
- Automated build, testing, and deployment workflows using Git and CI/CD tools, supporting consistent releases across development, staging, and production environments.

PROJECTS

Smart Chicken Disease Detection and Farm Management Solution **Aug 2024 – Nov 2024**

Venky's Kitchen and Farm https://smart-poultry-platform.vercel.app/

- Built a real-time poultry health platform that identifies sick, dead, and abnormally behaving birds from shed camera feeds,enabling early detection of potential outbreaks and reducing reliance on manual inspection.
- Designed a two-stage computer vision pipeline using Python and YOLOv8 for bird detection and a fine-tuned classification model for identifying indicators associated with coccidiosis, Newcastle disease, and Salmonella.
- Developed and deployed a Dockerized FastAPI inference service optimized for edge hardware and real-time processing.
- Automated poultry shed climate control using MQTT, integrating temperature, humidity, ammonia, and CO_2 sensors with automated fan, heater, and mister controls using hysteresis-based rules.

- Developed an operations dashboard using Next.js and Supabase/PostgreSQL, including real-time alerts, shed health scoring, mortality/outbreak reporting, role-based access, and attendance management.
- Implemented browser-based face check-in that runs client-side, ensuring worker images remain on the device.
- **Technologies:** Python, YOLOv8, PyTorch, OpenCV, FastAPI, MQTT, Docker, Next.js, Supabase, PostgreSQL, Edge AI, IoT, Real-Time Systems, RBAC.

AutoETL **Aug 2024 – Nov 2024**

Intisca Technologies https://autoetl-mu.vercel.app/

- Built a full-stack AI-powered ETL platform that converts natural-language instructions into executable ETL pipelines using an "LLM plans, engine executes" architecture.
- Designed the system so the LLM generates only schema-validated JSON DAGs rather than executable code, improving reliability and security.
- Developed a custom Python DAG execution engine using Polars and DuckDB with topological ordering, per-node retries, failure isolation, and post-run data-quality validation.
- Implemented real-time pipeline monitoring using Redis Pub/Sub and Server-Sent Events (SSE) to stream execution status to the browser.
- Added automated data-quality checks covering row-count changes, null-rate variations, and schema drift.
- Strengthened application security through Zod/Pydantic validation, sandboxed AST expression compilation, AES-256-GCM encrypted credentials, and PostgreSQL Row Level Security.
- Designed a provider-agnostic LLM layer with cost-aware model routing based on prompt complexity.
- Built a comprehensive automated test suite with 43 Python tests and 9 TypeScript tests.
- **Technologies**: TypeScript, Next.js 15, React Flow, Python, FastAPI, Polars, DuckDB, Redis, Supabase, PostgreSQL, Zod, Pydantic, LLM APIs, Docker.

EDUCATION

Master of Science in Data Science **Jan 2026**

University at Buffalo, The State University of New York, Buffalo, NY

Bachelor of Engineering – BE, Information Technology **Jun 2024**

Savitribai Phule Pune University, India

CERTIFICATION

Amazon Web Services (AWS)